

Solving Systems by Graphing

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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|--|---|
| 1. intersection point = $x = 2, y = 4$ | 7. intersection point = $x = 2, y = 6$ |
| 2. intersection point = $x = 2, y = 5$ | 8. intersection point = $x = 5, y = 4$ |
| 3. intersection point = $x = 3, y = 4$ | 9. intersection point = $x = 1, y = 5$ |
| 4. intersection point = $x = 3, y = 8$ | 10. (a) second equation solved for $y = 2x + 3$, — |
| 5. (a) solution, or none = $\emptyset$, — | 11. intersection point = $x = 3, y = 6$ |
| 6. intersection point = $x = 4, y = 4$ | 12. (a) intersection point = $x = 3, y = 9$, — |
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What is verified

12 of 15 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 5, 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.C.8a – problems 1, 2, 3, 4, 5, 6, 7, 8, 10, 11

8.EE.C.8c – problems 9, 12

Difficulty 1–4 of 5 across 15 tagged checks.

- 1.** $y = x + 2$ and $y = 6 - x$, both drawn. Read the intersection.

Solution: The lines meet two right and four up. Check: $4 = 2 + 2$ and $4 = 6 - 2$, both true. $(2, 4)$

- 2.** $y = 2x + 1$ and $y = 7 - x$, both drawn. Read the intersection.

Solution: They meet two right and five up. Check: $5 = 2(2) + 1$ and $5 = 7 - 2$. $(2, 5)$

- 3.** Graph $y = x + 1$ and $y = 10 - 2x$; give the intersection.

Solution: The first line starts at 1 and rises 1 for each 1 across; the second starts at 10 and falls 2 for each 1 across. They cross three right and four up, and $x + 1 = 10 - 2x$ confirms it: $3x = 9$, so $x = 3$ and $y = 4$. $(3, 4)$

- 4.** Graph $y = 2x + 2$ and $y = x + 5$; give the intersection.

Solution: Starting points 2 and 5, steps of 2 and 1. The lines meet at three across, eight up; algebraically $2x + 2 = x + 5$ gives $x = 3$ and then $y = 8$. $(3, 8)$

5. Graph $y = 3x + 1$ and $y = 3x + 4$. (a) Solution or none. (b) What the graphs show.

Solution (a): Both lines climb 3 for each 1 across, so they never meet. Setting $3x + 1 = 3x + 4$ leaves $1 = 4$, which is false. no solution

Solution (b) — instructor-judged. The two lines have the same steepness and different starting heights, so the vertical gap between them stays at 3 units the whole way across the grid. Lines that never meet give a system with no solution. Naming them parallel is only half the answer; the consequence for the system must be stated.

6. Graph $y = 8 - x$ and $y = \frac{1}{2}x + 2$; give the intersection.

Solution: The second line rises 1 for every 2 across, which is why its plotted points land on every second gridline. The lines meet four right and four up: $8 - x = \frac{1}{2}x + 2$ gives $6 = \frac{3}{2}x$, so $x = 4$ and $y = 4$. (4, 4)

7. Graph $x + y = 8$ and $y = 3x$; give the intersection.

Solution: For the first line, two easy points are $(0, 8)$ and $(8, 0)$. The second passes through the origin, rising 3 each step. They meet at two across, six up: $x + 3x = 8$ gives $x = 2$ and $y = 6$. (2, 6)

8. Graph $x = 5$ and $y = 9 - x$; give the intersection.

Solution: $x = 5$ is the vertical line through 5 on the x -axis — it has no slope-intercept form. The other line passes through $(0, 9)$ falling 1 each step, and at $x = 5$ it is at $y = 4$. (5, 4)

9. Courier $y = 3x + 2$ against rival $y = 5x$; give the point where the prices are equal.

Solution: The rival's line starts at the origin and is steeper, so it begins below and overtakes. They meet one across, five up: $3x + 2 = 5x$ gives $2x = 2$, so $x = 1$ and $y = 5$. (1, 5)

10. $y = 2x + 3$ and $4x - 2y = -6$. (a) Second equation solved for y . (b) How many solutions.

Solution (a): $4x - 2y = -6$. Subtract $4x$: $-2y = -6 - 4x$. Divide every term by -2 : $y = 3 + 2x$. $y = 2x + 3$

Solution (b) — instructor-judged. The second equation turns into the first, so the two graphs are the same line drawn twice. Every point on that line satisfies both equations, so the system has infinitely many solutions. A single ordered pair is wrong here, and “infinitely many” with no reason does not earn full credit — the student must say the two equations describe one line.

11. Graph $2x + y = 12$ and $y = x + 3$; give the intersection.

Solution: The first line passes through $(0, 12)$ and $(6, 0)$; within the grid, $(2, 8)$ and $(5, 2)$ are convenient. The lines meet three across, six up: $2x + (x + 3) = 12$ gives $3x = 9$, so $x = 3$ and $y = 6$. (3, 6)

12. Company M $y = x + 6$ against company N $y = 3x$. (a) Equal-price point. (b) Which is cheaper on a long trip.

Solution (a): $x + 6 = 3x$ gives $2x = 6$, so $x = 3$ and $y = 9$. (3, 9)

Solution (b) — instructor-judged. Beyond the crossing point, company N's line is

the higher of the two, so company M is cheaper for long trips; below it the order is reversed and company N is cheaper for short ones. The reason is the rate: company N charges 3 dollars a mile against company M's 1 dollar a mile, so N's line is steeper and eventually rises above M's. Naming a company with no graphical or rate-based reason does not earn full credit.